Kurapati Hari Praneeth

Generative AI Engineer | LLM Applications | RAG | Python
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SUMMARY

AI engineer with an internship at Navent AI building LLM and RAG systems in Python with FastAPI, LangGraph, and Pinecone: medical-code retrieval, rule-based coding validation, retrieval evaluation, and a real-time voice query system. Projects include a medicinal plant identification app (PyTorch, FastAPI, React) and a network intrusion detection system (scikit-learn, XGBoost, Flask).

SKILLS

Languages: Python
AI / ML: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, Hugging Face Transformers, Sentence-Transformers
Generative AI: LLMs, RAG, LangGraph, Prompt Engineering, LLM Evaluation, Ollama
Retrieval & Data: Pinecone, FAISS, PostgreSQL
Backend & Web: FastAPI, Flask, REST APIs, JWT, React, AWS (S3, STS, SQS, Chime, Polly, Transcribe), Docker, Git, GitHub

EXPERIENCE

AI Engineering Intern — Breviq AI | Hyderabad, India Jun 2026 – Present
Navent AI — Clinical AI Platform
Python, FastAPI, LangGraph, Pinecone, PostgreSQL, AWS S3, AWS Chime, LLMs

- Built clinical documentation workflows in Python, FastAPI, and LangGraph that pair RAG retrieval with LLM analysis across 5 healthcare domains.
- Built an embedding benchmark and migrated clinical-code retrieval from ClinicalBERT to BGE on Pinecone; CDT Recall@10 increased from 61.3% to 84.7% while embedding storage decreased 69%.
- Implemented deterministic Python validators for medical, dental, and physical-therapy coding, including 12 medical inference guards, negation checks, dental billing rules, and CMS 8-minute validation.
- Built Navent Voice, a real-time CDI query resolution system integrating AWS Chime, Transcribe, Polly, and LLMs with conversational state management and safety guardrails.
- Extended the FastAPI backend with event-loop offloading, S3/STS storage, SQS batch processing, PostgreSQL persistence, and document merging.

PROJECTS

MediPlant AI — Medicinal Plant Identification & RAG Assistant Jun 2025 – Nov 2025
Python, PyTorch, FastAPI, FAISS, Sentence-Transformers, Ollama, React

- Built a medicinal plant identification app: a MobileNetV3-Small classifier identifies the plant, and a local Ollama LLM answers plant-specific questions with source citations.
- Trained MobileNetV3-Small on 6,884 images across 78 species (93.9% test accuracy, 0.94 macro F1). Added background augmentation, which raised accuracy under background shift from 43.5% to 77.6%.
- Built an entity-aware FAISS retrieval pipeline with MiniLM embeddings and plant-specific reranking; on an 89-question benchmark it reached 100% Hit@1 for plant retrieval and 100% Hit@5 for answer-chunk recall.

Enhancing Data Transmission Security in Cloud Using Machine Learning Dec 2025 – Mar 2026
Python, Flask, scikit-learn, XGBoost, TensorFlow/Keras, Docker

- Built a Flask network intrusion detection system on the NSL-KDD dataset (125,973 records, 41 features), classifying traffic into 5 classes.
- Compared 5 ML models: XGBoost scored 99.99% accuracy and 99.99% macro F1, ahead of Random Forest and the ANN-based models.
- Added AES-256-CBC encryption, JWT-authenticated REST APIs, MinMaxScaler preprocessing, and Docker packaging for model serving.

EDUCATION

B.Tech in Information Technology — St. Peter's Engineering College, Hyderabad, India 2022 – 2026
CGPA: 7.0/10

CERTIFICATIONS

- AI Engineer Associate Certificate Course — Udemy | Sep 2026
- Machine Learning & AI Foundations Course — Data Science Academy | Udemy | Sep 2026